

HRRP Coordination Meeting, Kavre

NRA Hall

30th May 2017

DUDBC Design Catalogue Vol. 2

- DUDBC has prepared the document.
- It covers seventeen model design following twelve technologies.
- The designs provided in this catalogue are based on calculations, model test and analytical tests as these technologies are not covered by Nepal National Building Code, 2060.
- It covers mainly Alternative Construction Materials and Technologies.

Catalogue Objective

To pave way for use of alternate materials and technologies in the reconstruction process as per the principles set by Post Disaster Needs Assessment (PDNA) for housing and human settlements recovery and reconstruction.

Catalogue Limitation

Designs included in this catalogue can be selected and used as they are, for reconstruction of urban and rural housing . For designs, other than those included in this catalogue, detailed engineering design and approval from concerned authorities shall be done.

Lists of Model

- 1. Interlocking Brick Masonry One Storey**
 - One Storey
 - One Storey
 - Two Storey
- 2. Confined Hollow Concrete Block Masonry**
 - Two Storey
- 3. Hollow Concrete Block Masonry**
 - Two Storey
- 4. Compressed Stabilized Earth Block Masonry**
 - One Storey
 - Two Storey
- 5. Random Rubble Masonry in Mud mortar with GI wire Containment**
 - One Storey
 - Two Storey
- 6. Bamboo and Stone Masonry Hybrid Structure**
 - Two Storey
- 7. Rat Trap Bond Masonry**
 - One Storey
- 8. Earth Bag Masonry**
 - One Storey
- 9. Light Gauge Steel Structure**
 - One Storey
 - Two Storey
- 10. Steel Structure**
 - Two Storey
- 11. Timber Structure**
 - Two Storey
- 12. Debris Block Masonry**
 - Two Storey

Corrections & Exceptions Manual

Many houses have been found during inspection to be non-compliant with MRs. These buildings are therefore vulnerable to future earthquakes, as they do not have the required earthquake resistant elements. In order to bring these houses to compliance, corrective measures are required

Limitation

This corrections / exceptions manual covers only load bearing masonry buildings under the GoN housing reconstruction programme. The corrective measures are only for buildings that are newly constructed, or under construction.

Organization of the Manual

- Minimum requirements
- Problem statements
- Common defects
- Exceptions
- Solutions

SITE CONDITION

HAZARD AREA	Exception / Correction
Geological fault or Ruptured area	- Not to be used for residential building
Water-Logged area River Bank	- Construction is allowable if the site is appropriately treated. - Maintain minimum distance from river bank and observed high flood level.
Steep Slope	- If the terrain is stable and soil is medium to hard, construction on steep slopes is allowable.
Filled Area	- If a building is to be constructed on filled-ground, the foundation shall be deep enough so as to rest on the firm ground surface beneath the fill.
Rock-fall Area	- Building can be constructed in such areas except in risky rock fall area identified by geological study and local knowledge.

Review of Typical Non-Compliant RCC Structures

- Cantilever 1.5 m long
- 1x2 , 1x3, 1x4 bays
- 1x1 bay, two side eccentric footing
- Room size (4.5mx4.5m) & (5.5mx3.5m)
- Column at Centre of grid missing
- Passage 1.5m
- Footings at differential level
- Column size 9"x12"